Self-Critique Without Structured Gain: Residual-Stream Drift Under Reflection Mirrors Re-Decoding

Ankit Singh and NeuriCo

Abstract

Self-critique prompting is widely used to improve large language model reasoning, but it is unclear whether it produces meaningful internal state updates or only a second sample from the same policy. We study this question by jointly measuring behavior and residual-stream representation drift under a controlled evaluation protocol. On 48 examples (24 GSM8K and 24 CommonsenseQA), we compare four conditions with QWEN2.5-1.5B-INSTRUCT: one-shot baseline, re-decode control, intrinsic self-critique revision, and an external-critique subset using GPT-4.1. We extract per-layer final-token residual representations and quantify drift with cosine distance and L2 displacement, then test condition differences with paired non-parametric statistics and FDR correction.

Self-critique substantially reduced accuracy from 39.6% to 16.7% (delta -22.9 points, 95% CI $[-37.5, -6.25]$, McNemar $p = 0.0098$). Re-decoding also reduced accuracy to 25.0%, indicating broad second-pass instability. Critically, mean cosine drift was nearly identical for self-critique and re-decode (0.5700 vs 0.5719), and layer-wise differences were non-significant after Benjamini–Hochberg correction across 28 layers. External critique did not recover performance on its subset.

These results provide no evidence, in this setup, for uniquely structured reflection-induced residual drift. Instead, self-reflection behaves closer to destabilizing regeneration than targeted internal correction.

1 Introduction

Prompting a language model to critique and revise its own answer is now common in research systems and production agents. The core assumption is that reflection improves reasoning quality, not only output form. Our main question is simple: does self-critique induce structured internal representation change, or does it mostly trigger another unstable decode?

Why this matters. If reflection truly reshapes internal states, it supports mechanistic claims about iterative reasoning and safer recursive agent loops. If it mostly re-samples outputs, then observed gains in some settings may be fragile prompt effects rather than robust internal correction. Distinguishing these regimes requires coupling output-level metrics with representation-level measurements.

What is missing in prior work? Reflection literature reports mixed behavioral findings. Iterative pipelines such as Self-Refine and Reflexion often show gains in selected settings [Madaan et al., 2023, Shinn et al., 2023], while other studies show that intrinsic self-correction can fail without stronger feedback or grounding [Huang et al., 2024, Gou et al., 2024, Chen et al., 2024]. In parallel, representation studies show that high-level behavior can align with low-dimensional activation structure [Zou et al., 2023, Turner et al., 2023, Zhao et al., 2024, Arditi et al., 2024]. However, few works jointly test reflection behavior and residual-stream drift under one controlled protocol.

Our approach. We evaluate QWEN2.5-1.5B-INSTRUCT on balanced subsets of GSM8K and CommonsenseQA with deterministic decoding. We compare baseline answers, a re-decode control, intrinsic self-critique revision, and a small external-critique condition. We then measure per-layer residual drift from baseline to revised trajectories and test whether self-critique causes larger or more structured movement than re-decoding.

Quantitative preview. Self-critique reduced accuracy from 39.6% to 16.7% (McNemar $p = 0.0098$), while re-decoding reduced it to 25.0%. Mean cosine drift under self-critique and re-decoding was nearly identical (0.5700 vs 0.5719), with no layer surviving FDR correction.

Contributions.

- We propose a unified protocol that pairs reflection behavior with residual-stream drift analysis on the same examples.
- We conduct a controlled comparison between intrinsic self-critique and non-critique re-decoding to isolate second-pass effects.
- We provide statistical evidence that, in this model-task-prompt regime, self-critique does not yield unique structured drift and instead degrades performance.
- We complement headline metrics with transition counts, layer-wise tests, and probe diagnostics that expose underpowered correction separability.

The rest of the paper reviews related work (section 2), details the protocol (section 3), reports results (section 4), and discusses limitations and implications (section 5).

2 Related Work

Self-reflection and iterative revision. Methods such as Self-Refine and Reflexion show that language-only critique loops can improve outputs in many tasks [Madaan et al., 2023, Shinn et al., 2023]. These papers motivate reflection as a practical control strategy, but they focus on outcome metrics and do not directly test internal representation geometry.

When intrinsic correction fails. Several studies report that models often fail to detect their own reasoning errors, and that performance can degrade after self-correction unless stronger signals are provided [Huang et al., 2024, Chen et al., 2024]. Tool-grounded critique (for example, CRITIC) can improve reliability by adding external verification [Gou et al., 2024]. Recent counter-evidence also argues that intrinsic correction can succeed under carefully controlled prompting and decoding conditions [Liu et al., 2024]. Our work does not attempt to resolve all setting-dependent disagreements; instead, we test whether representation drift can distinguish critique-conditioned revision from generic second-pass generation.

Representation geometry and activation-level control. Representation engineering and activation steering show that behavior is often encoded in identifiable directions or subspaces [Zou et al., 2023, Turner et al., 2023, Arditi et al., 2024]. Residual-stream analyses under knowledge conflict further demonstrate that intermediate states can predict response behavior [Zhao et al., 2024]. Building on this line, we focus on residual drift under reflection and explicitly compare it against a re-decode control to test whether self-critique creates a distinct geometric signature.

Positioning. Unlike prior reflection papers that stop at output gains, we jointly evaluate behavior, per-layer drift, and correction separability. Unlike prior representation papers that do not include reflection loops, we evaluate how critique-conditioned trajectories move relative to baseline and relative to a non-critique second attempt.

3 Methodology

Problem setup. For each question x with gold label y, we generate answers under four conditions: baseline ($a^{(0)}$), re-decode control ($a^{(r)}$), intrinsic self-critique revision ($a^{(s)}$), and external-critique revision ($a^{(e)}$) on a subset. We evaluate both behavioral change and internal representation drift relative to baseline.

Table 1: Condition-level accuracy on the 48-example evaluation set. Best result is bolded.

Condition	Accuracy
Baseline	**0.3958**
Re-decode control	0.2500
Self-critique revised	0.1667
External-critique revised (subset, n=16)	0.2500

Table 2: Dataset-wise accuracy by condition. Best value in each row is bolded.

Dataset	Baseline	Re-decode	Self-critique
COMMONSENSEQA (n=24)	**0.7917**	0.5000	0.2500
GSM8K (n=24)	0.0000	0.0000	**0.0833**

Datasets and sampling. We use 24 sampled examples from GSM8K test and 24 sampled examples from CommonsenseQA validation (seed 42), yielding 48 total examples with balanced task composition. GSM8K scoring parses numeric final answers; CommonsenseQA scoring parses final option letters A–E.

Models and decoding. The local model is QWEN2.5-1.5B-INSTRUCT (1.5B Instruct). External critique uses GPT-4.1 for 16 examples due to runtime and API cost constraints. Decoding uses temperature 0.0 and max 96 new tokens.

Representation extraction. Let $\boldsymbol{h}_{i,\ell}^{(c)}$ be the final-token residual representation for example i, layer ℓ, and condition $c \in \{r, s, e\}$. We compute cosine drift from baseline

$$D_{i,\ell}^{\cos}(c) = 1 - \frac{\boldsymbol{h}_{i,\ell}^{(0)} \cdot \boldsymbol{h}_{i,\ell}^{(c)}}{\|\boldsymbol{h}_{i,\ell}^{(0)}\| \, \|\boldsymbol{h}_{i,\ell}^{(c)}\|}, \tag{1}$$

and L2 drift $D_{i,\ell}^{\ell_2}(c) = \|\boldsymbol{h}_{i,\ell}^{(c)} - \boldsymbol{h}_{i,\ell}^{(0)}\|_2$.

Statistical analysis. We report accuracy deltas with bootstrap 95% CIs, McNemar tests for paired correctness changes, and paired Wilcoxon signed-rank tests for layer-wise drift differences between self-critique and re-decode, corrected with Benjamini–Hochberg FDR across 28 layers.

Probe analysis. We train logistic-regression probes on drift profiles to classify whether initially incorrect examples are corrected after self-critique. Because only 2 of 29 probe examples are positive, we treat probe metrics as diagnostic rather than confirmatory.

Implementation details. We run Python 3.12.8 with torch 2.10.0+cu128, transformers 5.3.0, datasets 4.6.1, scikit-learn 1.8.0, scipy 1.17.1, and statsmodels 0.14.6. Experiments were executed on two RTX 3090 GPUs with mixed-precision autocast and sequential generation, taking about 13 minutes for the main 48-example run.

4 Results

Main outcomes. Table 1 shows that self-critique performs worst among full-set conditions. Accuracy drops from 0.3958 to 0.1667 (↓), while re-decoding also drops to 0.2500. The external-critique subset reaches 0.2500 and does not recover baseline subset performance.

Dataset-level behavior. Table 2 shows that baseline performance is carried mainly by CommonsenseQA (0.7917), where both second-pass conditions degrade strongly. GSM8K is near-floor for baseline and re-decode, with a small absolute increase under self-critique (0.0833), but this does not offset the larger CSQA degradation.

Paired significance and transitions. Self-critique causes a significant paired drop (McNemar $p = 0.0098$). Transition counts from baseline to self-revised are 2 improved, 13 degraded, and 33 unchanged, confirming that harmful revisions dominate.

Table 3: Key statistical tests comparing reflection conditions against baseline or control.

Metric	Value
Self - Baseline delta	-0.2292 (95% CI: $[-0.3750, -0.0625]$)
Re-decode - Baseline delta	-0.1458 (95% CI: $[-0.2708, -0.0208]$)
McNemar p (Self vs Baseline)	0.0098
Mean cosine drift (Self)	0.5700
Mean cosine drift (Re-decode)	0.5719
Paired Wilcoxon p (mean drift, Self vs Re-decode)	0.7643
Layer-wise FDR-significant differences (28 tests)	0

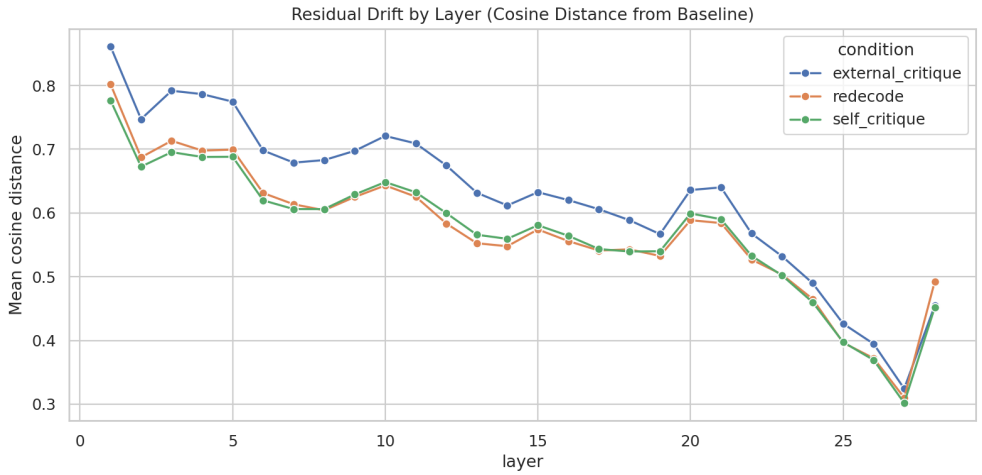

Figure 1: Layer-wise cosine drift from baseline hidden states. Self-critique and re-decode have highly similar profiles across all 28 layers, with no FDR-significant layer-wise difference.

Representation drift. Figure 1 plots per-layer cosine drift profiles. Self-critique and re-decode curves closely overlap across layers, and no layer remains significant after FDR correction (28 tests). On per-example mean drift, paired Wilcoxon $p = 0.7643$, consistent with no measurable critique-specific increase in drift magnitude.

Accuracy and correction visuals. Figure 2 summarizes condition-level accuracy, and figure 3 shows sparse correction outcomes among initially wrong examples.

Probe separability. Probe evaluation is underpowered: only 2/29 positives (6.9%) among initially wrong examples. Several folds produce undefined AUC, and mean F1 is 0.0. We therefore avoid strong claims about separability in this run.

5 Discussion

What do the results imply? In this setup, self-critique behaves more like destabilizing regeneration than structured correction. Behavioral degradation is large and statistically supported, while internal drift magnitude is almost indistinguishable from a non-critique second decode.

Why might this happen? First, critique content is often generic (for example, "double-check" style feedback) and may fail to localize concrete reasoning errors, consistent with prior decomposition studies [Chen et al., 2024]. Second, baseline GSM8K performance is near zero, suggesting model capability limits that critique prompts alone cannot overcome. Third, deterministic second-pass generation may amplify prompt-format brittleness instead of improving reasoning.

Limitations. This study uses one local model, one prompt family, and a moderate sample size (48 examples). External critique is evaluated on only 16 examples. Representation analysis uses final-token layer states rather than full token trajectories, and probe analysis is heavily class-imbalanced.

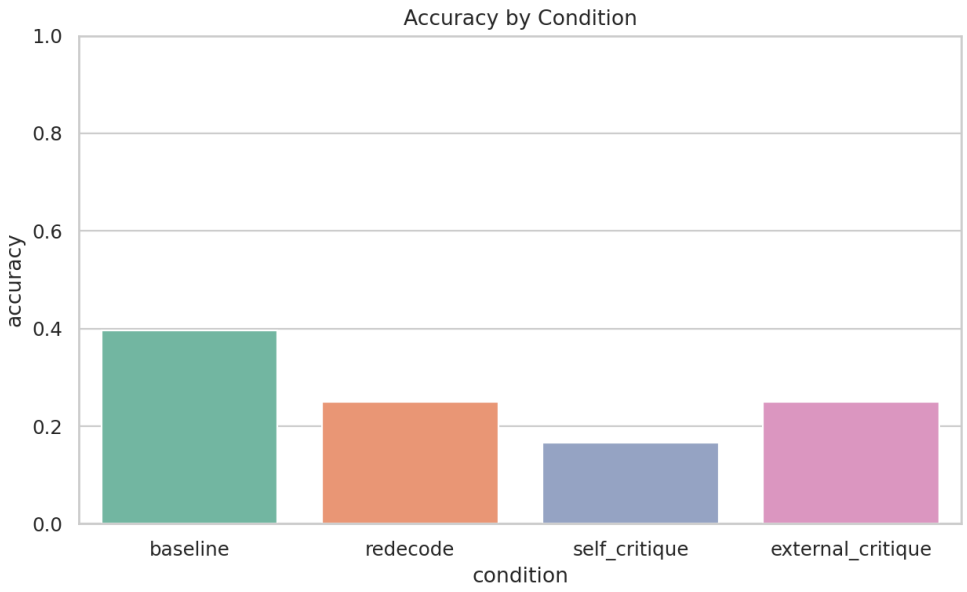

Figure 2: Accuracy by condition. Both second-pass strategies reduce performance relative to one-shot baseline, with the largest drop under intrinsic self-critique.

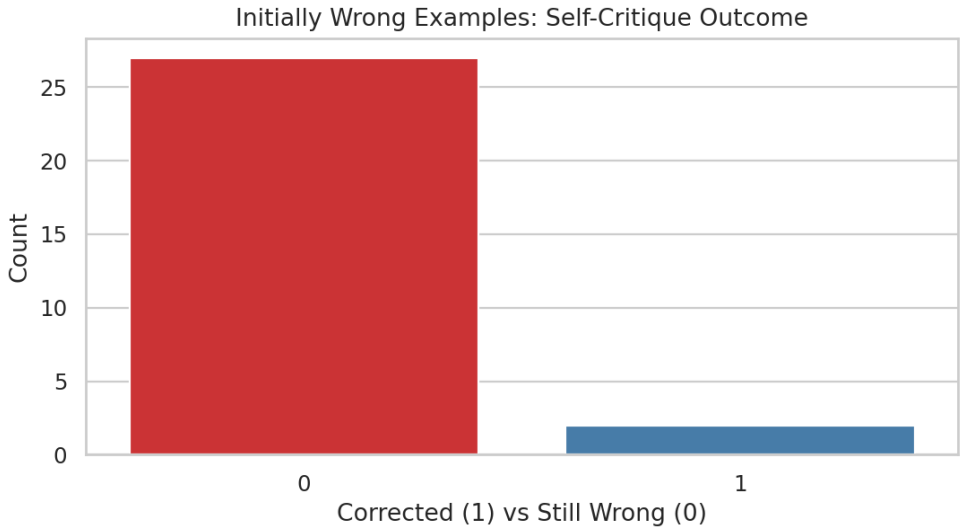

Figure 3: Correction outcomes for self-critique transitions. Only two initially wrong examples are corrected, while many initially correct cases degrade.

Broader implications. Reflection prompting should not be assumed helpful by default, especially in low-capacity or mismatched regimes. For interpretability, apparent output changes after critique do not necessarily indicate unique internal-state shaping. More reliable reflection may require explicit error localization, tool grounding, or stronger base models.

6 Conclusion

We presented a controlled evaluation of self-critique through both behavior and residual-stream geometry. On balanced GSM8K and CommonsenseQA subsets with QWEN2.5-1.5B-INSTRUCT, in-

trinsic self-critique reduced accuracy by 22.9 absolute points relative to baseline and did not produce larger or more structured residual drift than a re-decode control.

The main takeaway is that reflection prompts, in this regime, did not induce distinct internal correction dynamics; they mostly triggered a harmful second-pass rewrite process.

Future work should test stronger local models (7B+), add explicit error-localization scaffolds, and analyze token-level trajectories with CKA/CCA-style subspace metrics. A scaling study across model sizes and critique sources (self, external model, and human) is a direct next step for identifying when reflection becomes genuinely state-shaping.

References

Andy Arditi, Oscar Obeso, Amir Syed, et al. Refusal in language models is mediated by a single direction. *arXiv preprint arXiv:2406.11717*, 2024.

Xinyun Chen, Yizhong Wang, Denny Zhou, et al. Large language models cannot find reasoning errors, but can correct them given error location. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.

Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yiming Yang, Nan Duan, and Weizhu Chen. Critic: Large language models can self-correct with tool-interactive critiquing. *International Conference on Learning Representations*, 2024.

Jie Huang, Kevin Chang, Wenhao Wang, Shayne Longpre, Akari Asai, Weijia Wang, et al. Large language models cannot self-correct reasoning yet. *International Conference on Learning Representations*, 2024.

Chao Liu, Han Li, Yuxuan Lai, Yujia Qin, Yao Lu, et al. Large language models have intrinsic self-correction ability. *arXiv preprint arXiv:2406.15673*, 2024.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. Self-refine: Iterative refinement with self-feedback. *arXiv preprint arXiv:2303.17651*, 2023.

Noah Shinn, Beck Labash, Ashwin Gopinath, Andrew Critch, Shunyu Yao, and Karthik Narasimhan. Reflexion: Language agents with verbal reinforcement learning. *Advances in Neural Information Processing Systems*, 2023.

Alexander Turner, Ludwig Thiergart, Gavin Leech, et al. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.

Yuyang Zhao, Zhengxuan Wu, Atticus Geiger, et al. Analysing the residual stream of language models under knowledge conflicts. *arXiv preprint arXiv:2410.16090*, 2024.

Andy Zou, Zifan Wang, Nicholas Carlini, et al. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023.